

Externa Biosciences — \$3.0M pre-seed

Cell-free platform ("QuickCharge") that inserts *any* non-standard amino acid into *any* protein at high throughput, bypassing the cell engineering that constrains every incumbent. The technology claim is unusually well-evidenced for pre-seed; the round buys three *in vivo* hypotheses, none of which is yet proven.

VERDICT

CONDITIONAL — 6/10

strongest of the three decks; invest gated on the diligence list below

WHAT YOU ARE BUYING

A tool becoming a drug company

\$4M non-dilutive revenue already funded the platform — real external validation

THE DECISIVE RISK

Business-model ambiguity

tools/services economics vs therapeutic-platform valuation; the round chooses therapeutics

KILL CONDITION

All three programs fail *in vivo*

15-month runway to xenograft/PK readouts — a genuinely binary seed gate

What they claim, and how well it is evidenced

CLAIM	EVIDENCE IN DECK	GRADE
Any nsAA into any protein, peptides → IgGs	100+ nsAAs across 12 categories, 7 scaffolds, up to 16 sites, 3 distinct nsAAs per protein	Strong — breadth is the differentiator and it is quantified
1-step / ~2h / up to 90% yield vs 3-step / 5d / 1–5%	Process comparison table, "scaled in grams"	Plausible, unverified — no third-party replication shown
Peer-reviewed foundation	Nature Communications 2024 (fluorogenic nanosensors); bioRxiv 2026 (covalent design)	One peer-reviewed, one preprint — verify the preprint's status
Commercial pull	Two signed paid contracts; \$4M non-dilutive; 8 pharma logos mapped to needs	Real but thin — logos are interest, not revenue

Team	de Puig (CEO), Kuru (CSO), Moret (CTO); Church + Collins as scientific co-founders; 23 patents, >10k citations	Very strong for the science; unproven as operators
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The honest core: the platform-capability claim is the best-evidenced part of the deck, and it is the part most companies fake. Externa quantifies breadth (sites, categories, scaffold sizes) rather than asserting it.

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Where the deck is weakest — the questions to ask

- **The comparables are chemistry, not platform, exits.** Synthorx (\$2.5B) and Ambrx (\$2.0B) were bought for *assets* built on one nsAA, not for the ability to install many. Ozempic's two nsAAs are incidental to its GLP-1 biology. The deck's implicit "we do more, so we're worth more" does not follow: pharma paid for de-risked molecules, not chemical flexibility. *Ask: which of your three programs becomes a Synthorx?*
- **All three targets are undisclosed.** Undisclosed targets at pre-seed are normal but they make the biological risk unpriceable; the deck's own claim of "lowest biological risk" cannot be checked. *Ask under NDA: target identity, competitive landscape per program, and why isoform selectivity is unaddressed by existing antibodies.*
- **Sutro is the natural experiment and the deck under-weights it.** Sutro (STRO) is a public cell-free nsAA company with GMP at 4,500L, three clinical candidates, and ~\$300M market cap on ~\$9.7M quarterly revenue. That is the market's current price for "cell-free + limited nsAA diversity, in the clinic." Externa's bet is that diversity is what was missing. *Ask: what does Sutro's valuation history tell you about how pharma pays for this?*
- **Manufacturing and IP.** "Scaled in grams" is bench-adjacent; ADC and IgG programs need CMC at kilograms with regulatory-grade characterisation. Freedom-to-operate around tRNA charging chemistry is unaddressed in the deck.
- **Round construction.** \$3.0M against three parallel *in vivo* programs (0.8 / 0.9 / 1.3 by phase) is thin; xenograft plus in-animal PK for three programs typically consumes more. *Ask what gets cut first if costs run over, and what the actual runway is in months.*

Verdict and terms

Conditional yes at pre-seed size, 6/10. This is the only one of the three companies whose central technical claim is both specific and independently checkable, and whose team's credentials match the claim. The risk is not that the technology doesn't work — the evidence says it does — but that a superb *tool* gets valued as a therapeutics platform before any in-vivo proof exists.

Pre-investment diligence (do these before wiring)

- Read both papers in full; confirm the bioRxiv preprint's peer-review status and whether the covalent-design claim survived review.
- Reference the two paid contracts: scope, value, renewal, and whether the counterparty is buying material or a program.
- Under NDA: the three targets, and one independent scientific reference on whether the isoform-selectivity approach is differentiated versus current antibody engineering.
- Confirm cap table and that the \$4M non-dilutive revenue carries no IP encumbrance or grant-back.
- Ask for the runway model in months with the three-program cost stack, and the fallback if one program consumes the others' budget.

Structure

Pre-seed sizing only — treat as a lottery ticket on a seed-stage step-up, not a position. The value inflection is the first in-vivo superiority readout (months 9–15); that readout, not the platform breadth, is what a seed investor will price. If you invest, ask for information rights on the in-vivo data specifically.

Sources: Externa pre-seed deck (15pp, read in full). No independent verification performed — all claims above are deck-stated unless marked.